

A
Major Project Phase-1 Presentation
on
“AGRO MIRAI: Predict, Protect, Prosper.”

Presented by

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Abstract

AGRO MIRAI: This is an AI-driven smart agriculture system that integrates satellite, weather, soil, and field data to deliver hyper-localized, predictive farming insights. It enables accurate crop recommendations, water availability, efficient irrigation planning, and early risk detection through machine learning, transforming complex data into simple, actionable guidance. With a cost-effective design, multilingual and voice-enabled interface, the system empowers farmers to enhance productivity, optimize resources, and adopt sustainable, climate-resilient farming practices.

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1. Introduction

- Agriculture suffers from lack of data-driven decisions and unpredictable conditions.
- AI and satellite technologies enable smart and precision farming.
- The system integrates weather, field data, and water availability.
- It delivers intelligent, season-wise crop planning and resource optimization



Fig. 1: A person checking about a field

2. Objectives

1. To develop a smart agriculture system that provides crop recommendations based on location, weather, and soil data .
2. To integrate real-time weather data and historical field data for improved decision-making .
3. To implement predictive analysis for irrigation planning and crop health monitoring.
4. To recommend season-wise cropping patterns based on water availability and environmental conditions .
5. To design a simple, multilingual, voice enabled and cost-effective system accessible to farmers .

3. Problem Statement

Traditional farming suffers from poor data integration and delayed decision-making, leading to inefficient crop selection. This project proposes an AI-driven smart agriculture advisory system that analyzes field history, weather, satellite, and water data to provide hyper-localized, predictive, season-wise crop recommendations.



Fig. 2: Challenges faced by farmers

4. Scope of the Project

In Scope :

- AI-based system integrating NDVI, weather, soil, and farmer data for crop recommendation, irrigation planning, and disease prediction.
- Provides hyper-local, season-wise agricultural insights with multilingual and voice-enabled support in a scalable and cost-effective manner.

Out of Scope

- Does not include automation or direct control of farming hardware such as irrigation pumps, drones, or IoT sensors.
- Does not cover financial services, market price systems, or government-level agricultural policy decision-making.

5. Literature Survey

Sl No.	Title	Authors	Year	Summary	Gaps Identified	Gaps Addressed
1	Human-Centered AI Design for Agriculture Advisory.	P. Reddy, S. Sharma.	2026	Focus on usability and farmer interactions with AI.	Lacks advanced prediction and integration.	Combines usability with predictive and integrated advisory system.
2	Smart Farming Technologies.	Raj & Prahades waran.	2025	Uses IoT, AI and remote sensing for precision agriculture.	High cost, poor integration, limited rural usability.	Cost-effective, integrated system with minimal hardware dependency .

SI No.	Title	Author	Year	Summary	Gaps Identified	Gaps Addressed
3	Promotion of Smart Farming Technologies to Smallholders.	S. R. Vincent, K. S. Senthilkumar, and S. Natarajan.	2024	Focuses on adoption, training, and awareness of smart farming practices.	Lack of intelligent systems, language barriers, low accessibility.	AI-based advisory with multilingual and voice-enabled support.
4	Smart Agriculture: A Literature Review .	Disha Garg, Mansaf Alam.	2023	Uses IoT, AI/ML , and DL for crop monitoring, irrigation, and prediction.	No unified system, poor UI, no offline support, lack of explainability.	Integrated system with simple UI, basic offline capability, and explainable advisory.

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6. Domain

Core Domain:

The core domain of AGRO MIRAI is Artificial Intelligence and Machine Learning for Precision Agriculture. It focuses on analyzing agricultural data to generate intelligent predictions and smart farming recommendations.

Application Domain :

The application domain of AGRO MIRAI is Smart Agriculture. It is used in Farm Advisory Systems and It is used to support farmers with crop recommendations, irrigation planning, and crop health monitoring.

7. Proposed System

7.1 Architectural Diagram

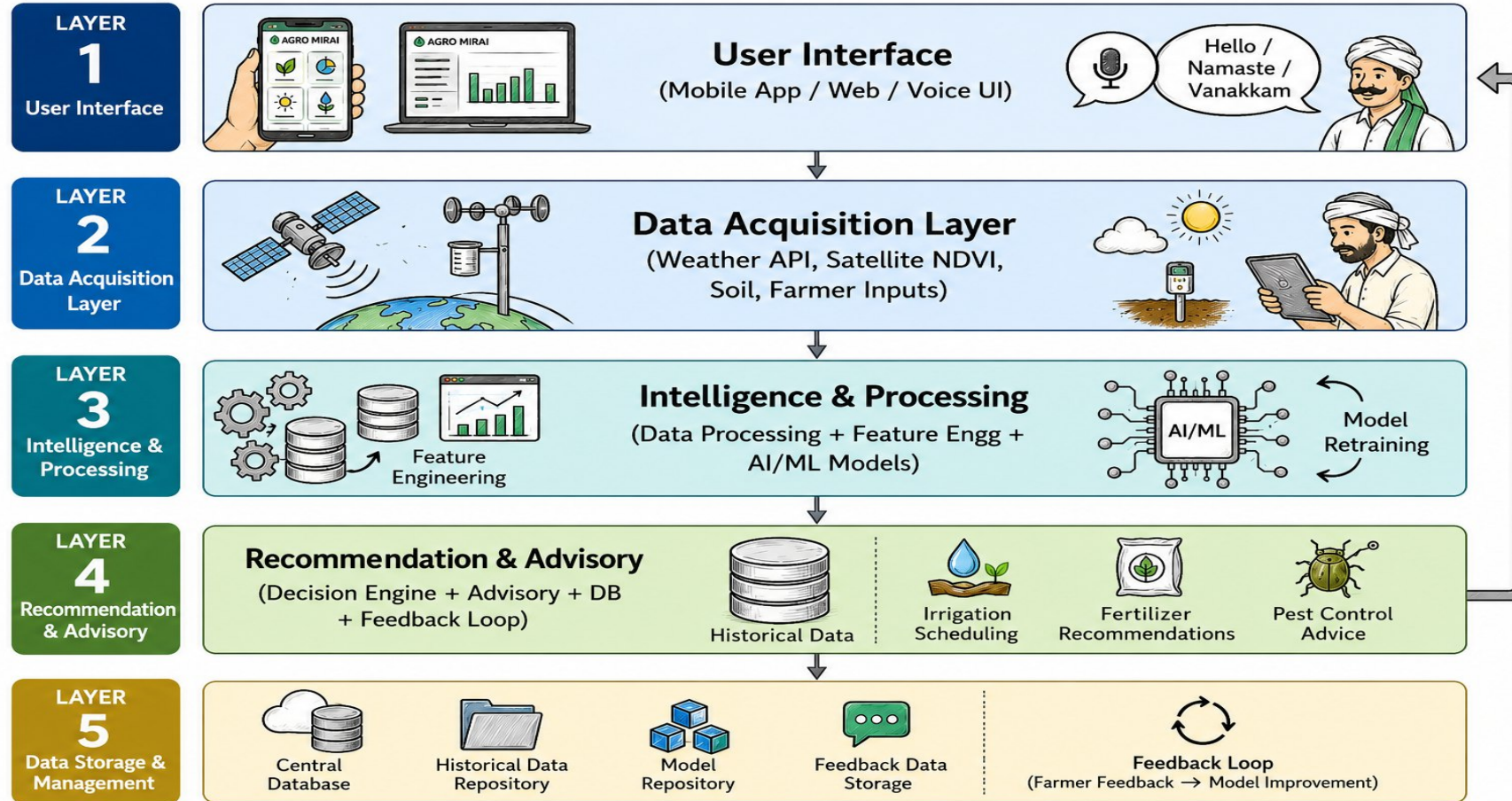


Fig 3: Layered System Architecture of the AGRO:MIRAI

7.2 Modules and its Functionality:

- 1. User Interaction Module** :Collects farmer inputs and provides multilingual, voice-based output.
- 2. Data Acquisition Module** :Gathers weather, satellite (NDVI), soil, and historical data.
- 3. Data Processing Module** :Cleans data and extracts important features for analysis.
- 4. AI/ML Prediction Module** :Predicts crop, irrigation needs, and disease risks.
- 5. Decision Support Module** :Gives final recommendations and alerts to farmers.

7.3 Block Diagram :

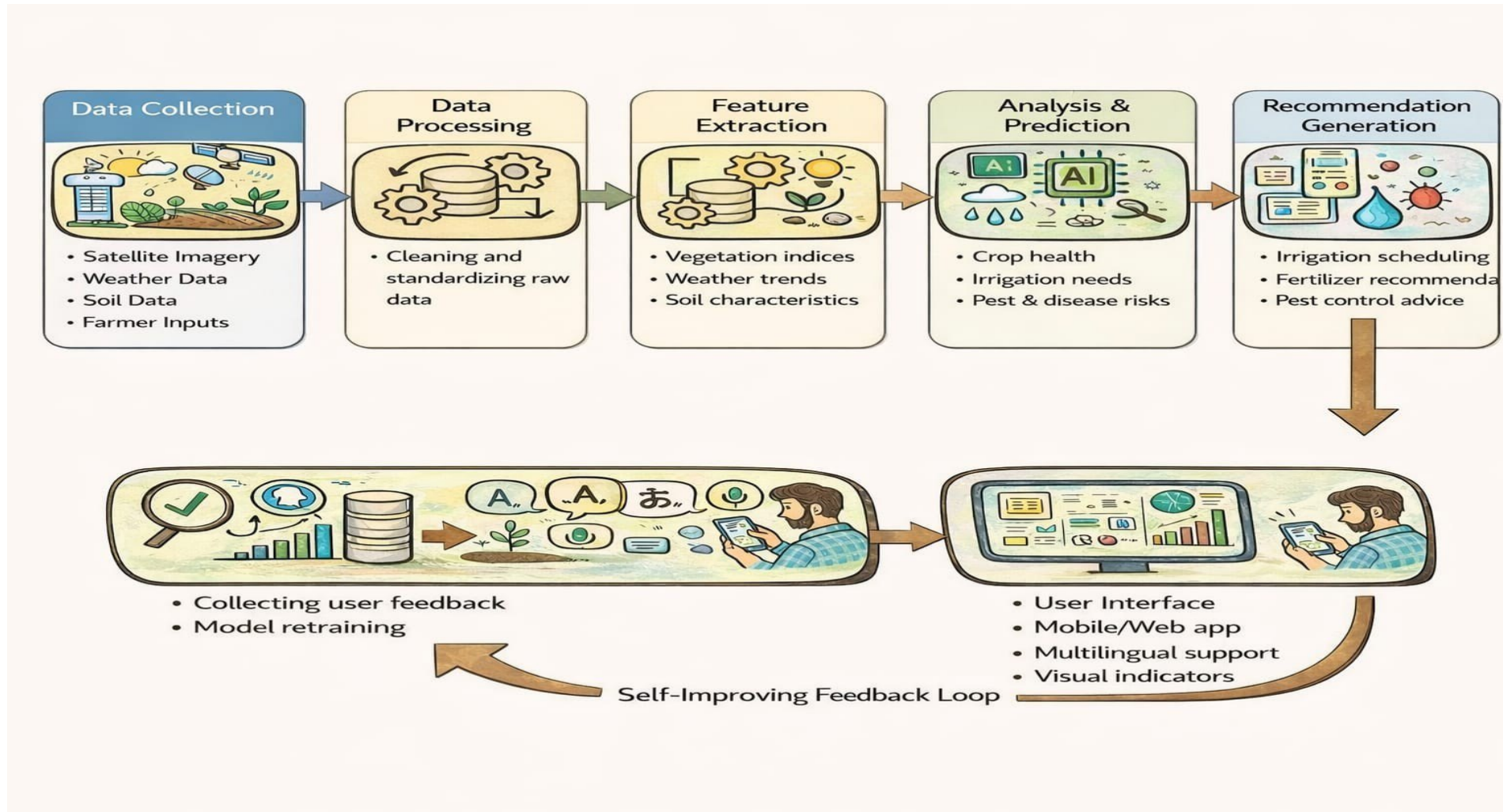


Fig 4: Block diagram of workflow

7.4 Improvement Over Existing Systems :

Ref. NO	Existing System	Proposed System
[1]	Uses single data source (weather/soil).	Integrates satellite, weather, soil, and field data.
[2]	Limited prediction capability.	AI-based predictive recommendation.
[3]	High cost & complex setup.	Cost-effective with minimal hardware.
[4]	Poor accessibility & usability.	Simple, multilingual, voice-enabled interface.
[5]	No explainability in results.	Provides explainable AI-based outputs.

8. Methodology

8.1 Techniques

- **Machine Learning Algorithms** – Random Forest, Gradient Boosting Decision Tree.
- **Regression Models** – Linear Regression, Random Forest Regression
- **Deep Learning** – Convolutional Neural Network(CNN) such as MobileNet.
- **Data Fusion** – Feature level data fusion & Data normalization
- **Spatio-Temporal Analysis** – NDVI, Location-based Clustering & seasonal trend Analysis .
- **Decision Support System** – Rule based inference, Alert generation and Threshold Analysis.
- **Explainable AI** – SHAP and feature importance analysis.

8.2 Development Model

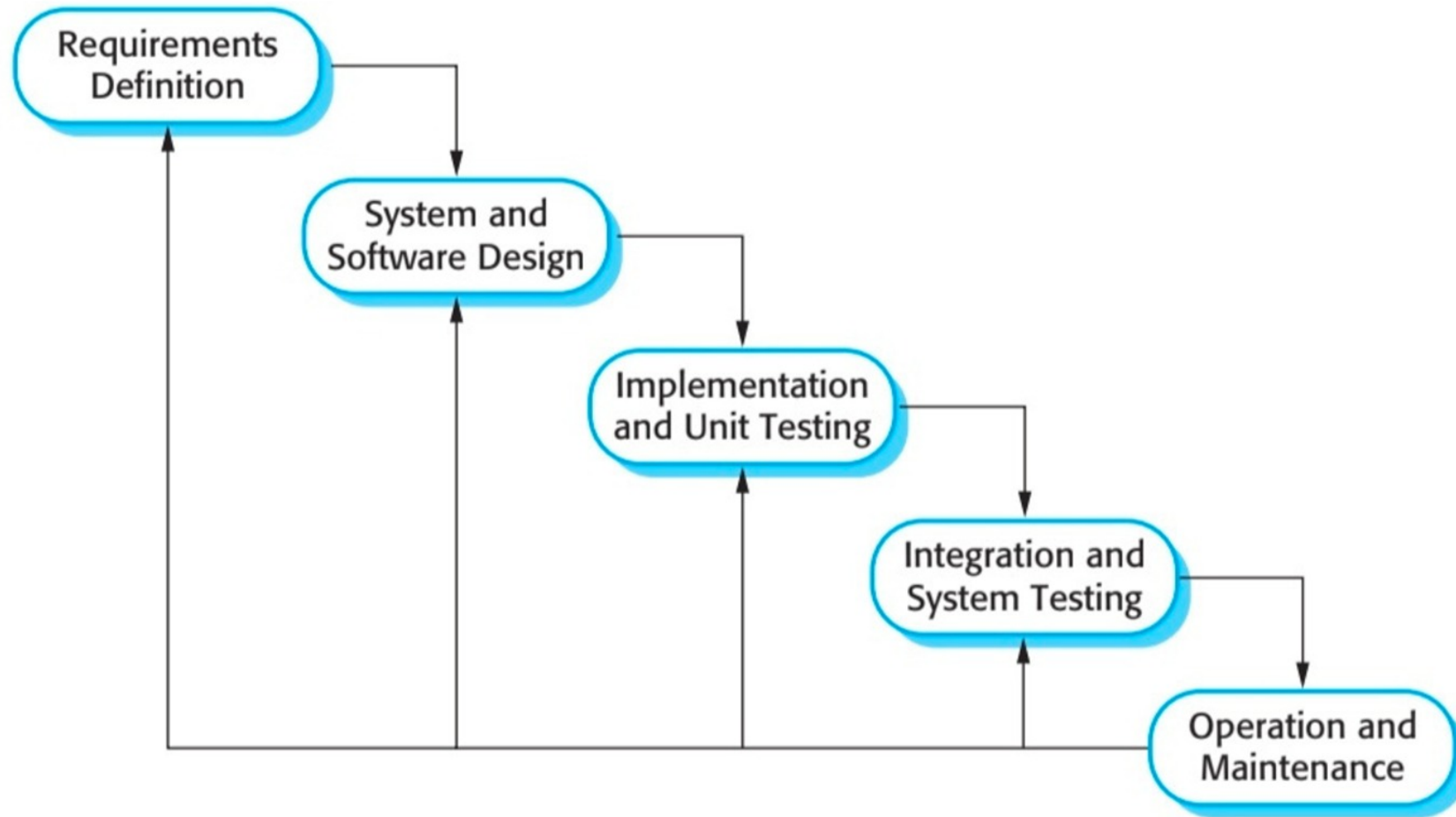


Fig 5: Waterfall Development Model for AGRO MIRAI

8.3 Tools And Technologies

- **Programming Languages:** Python.
- **Machine Learning Frameworks:** TensorFlow & Scikit-learn .
- **Data Processing:** Pandas, NumPy .
- **Weather Data API:** OpenWeatherMap API .
- **Satellite Data:** Google Earth Engine .
- **Database:** MySQL .
- **Backend Development:** Flask.
- **Frontend Development:** HTML, CSS, JavaScript .
- **Development Tools:** Visual Studio Code .

9. Requirements Specification

9.1 Functional Requirements

- Accept field, soil, weather, and location data .
- Provide crop recommendations.
- Generate irrigation suggestions.
- Predict crop health and disease risk .
- Produce alerts with explainable recommendations.

9.2 Non-Functional Requirements

- **Performance** – Provides fast and efficient advisory responses.
- **Scalability** – Supports increasing users and agricultural data.
- **Reliability** – Ensures consistent and accurate system operation.
- **Accessibility** – Enables easy access through multilingual interfaces.
- **Usability** – Provides a simple and farmer-friendly user experience.

9.3 Hardware Requirements

- **Processor:** Intel Core i3 or above .
- **RAM:** 4 GB minimum .
- **Storage:** 128 GB or above.
- **Display:** Standard monitor or laptop screen .

9.4 Software Requirements

- **Operating System:** Windows 10/11 or Linux .
- **Programming Language:** Python .
- **Frontend:** HTML, CSS, JavaScript .
- **Backend Framework:** Flask .
- **Database:** SQLite / MySQL .

10. Design

10.2.1 Context Diagram

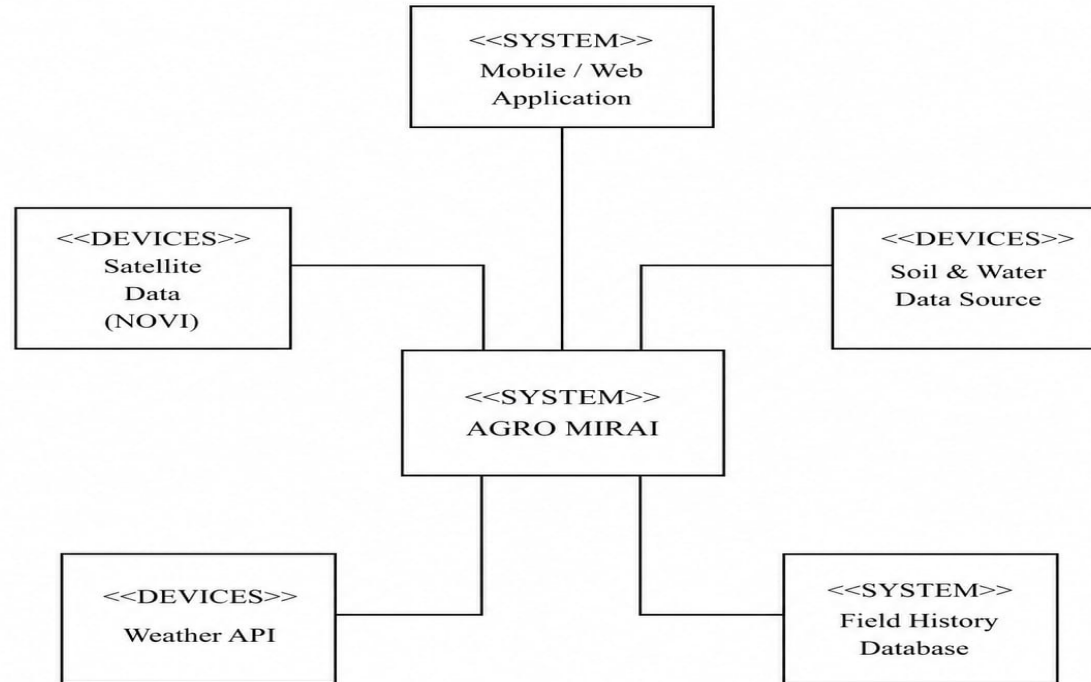
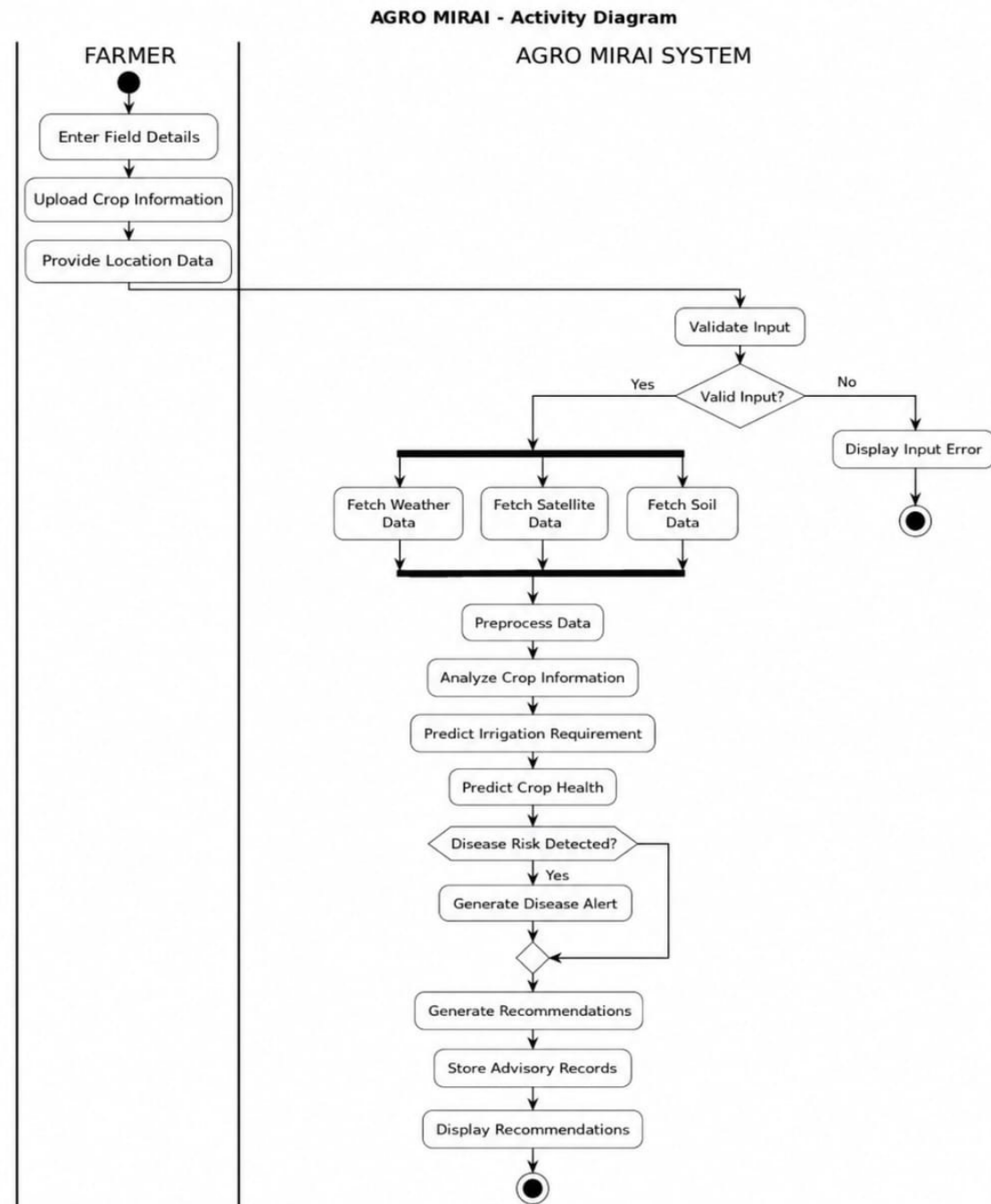


Fig 6: Context diagram of the AGRO MIRAI system.

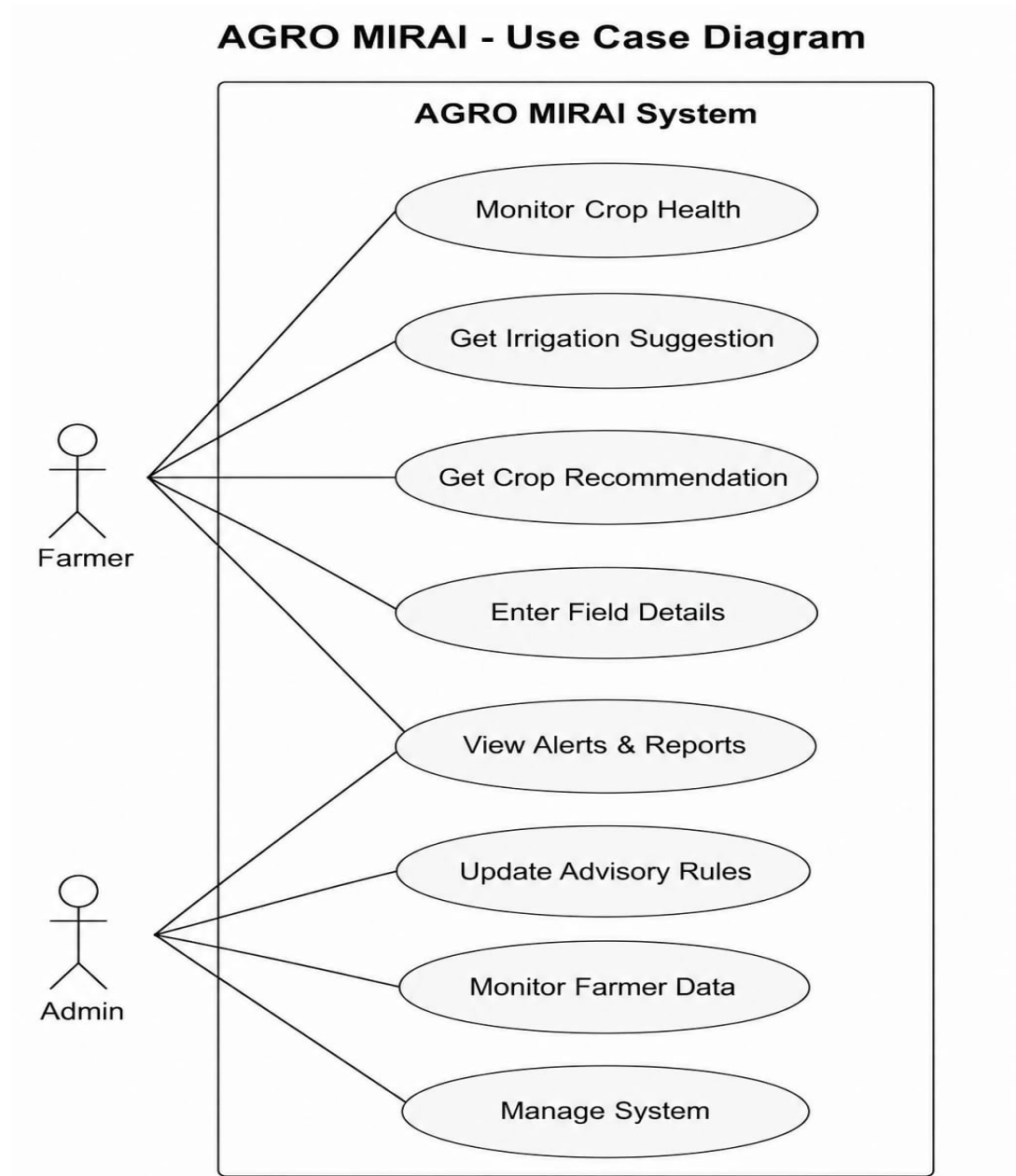
10.2.2 Activity Diagram

Fig 7: Activity diagram showing the system workflow.



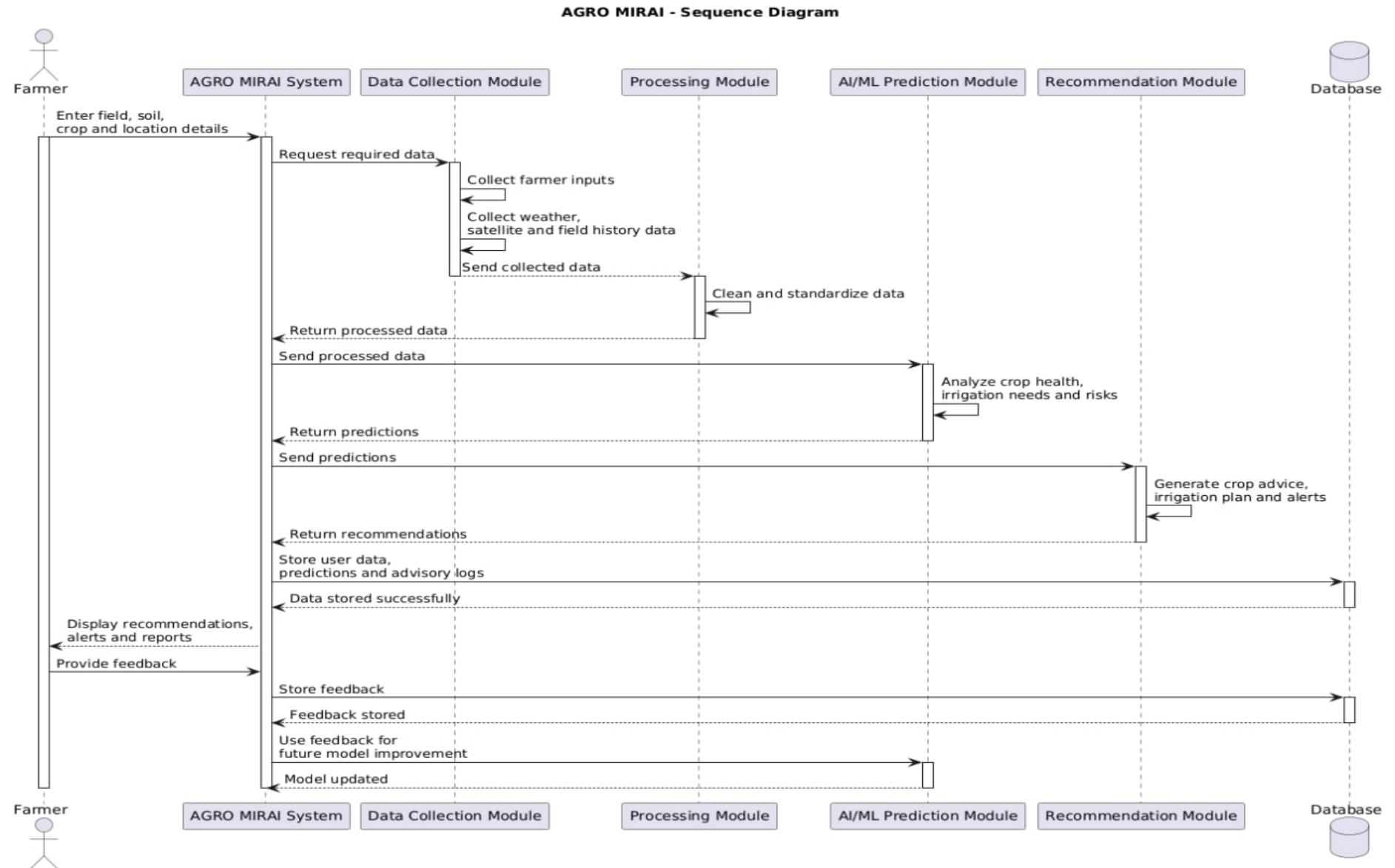
10.2.3 Use Case Diagram

Fig 8: Use case diagram representing user interactions.



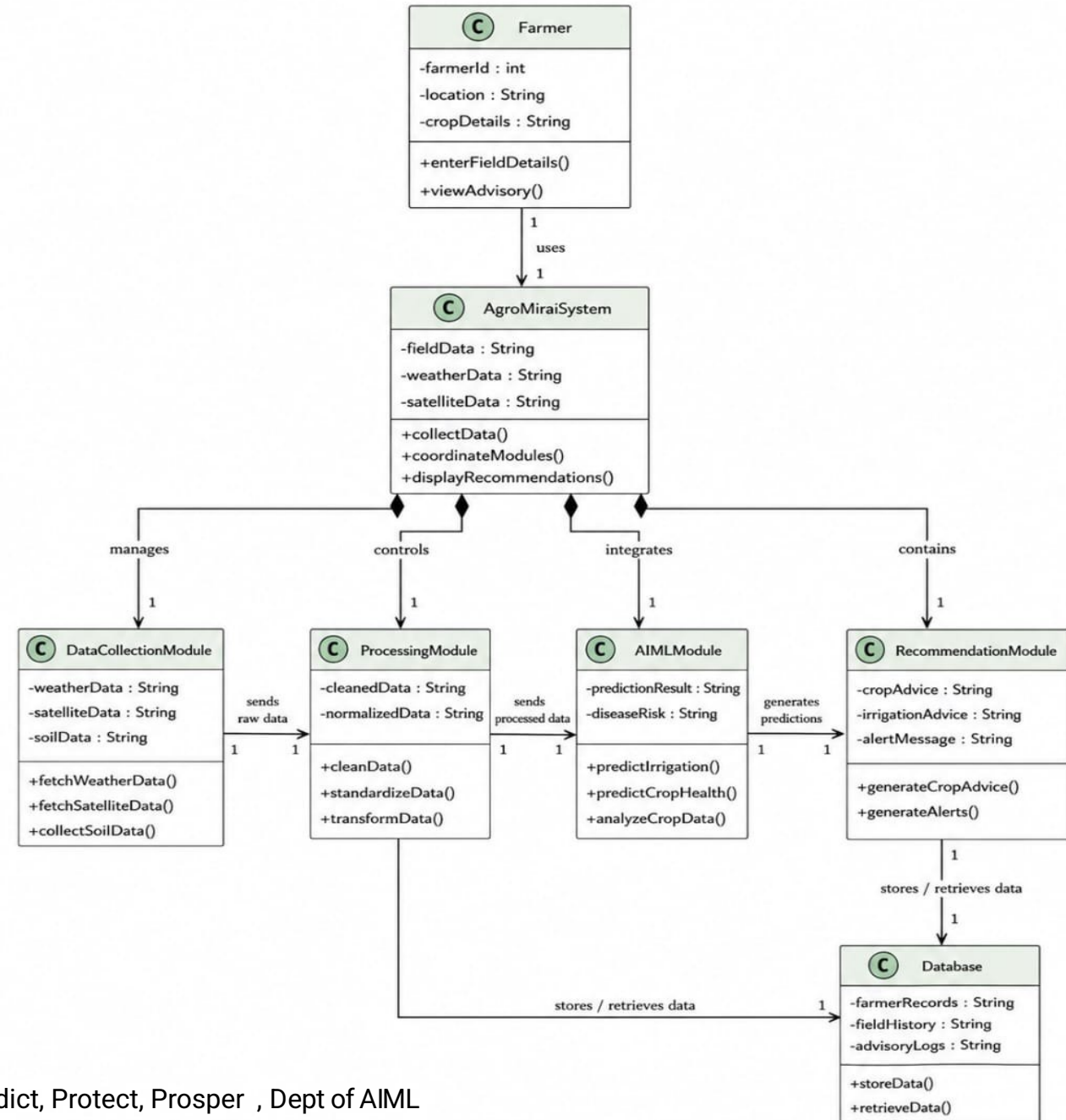
10.2.4 Sequence Diagram

Fig 9: Sequence diagram illustrating communication flow.



10.2.5 Class Diagram

Fig 10: Class diagram showing module relationships.



11. Expected Outcomes

- Accurate crop recommendations based on location, soil, and weather conditions

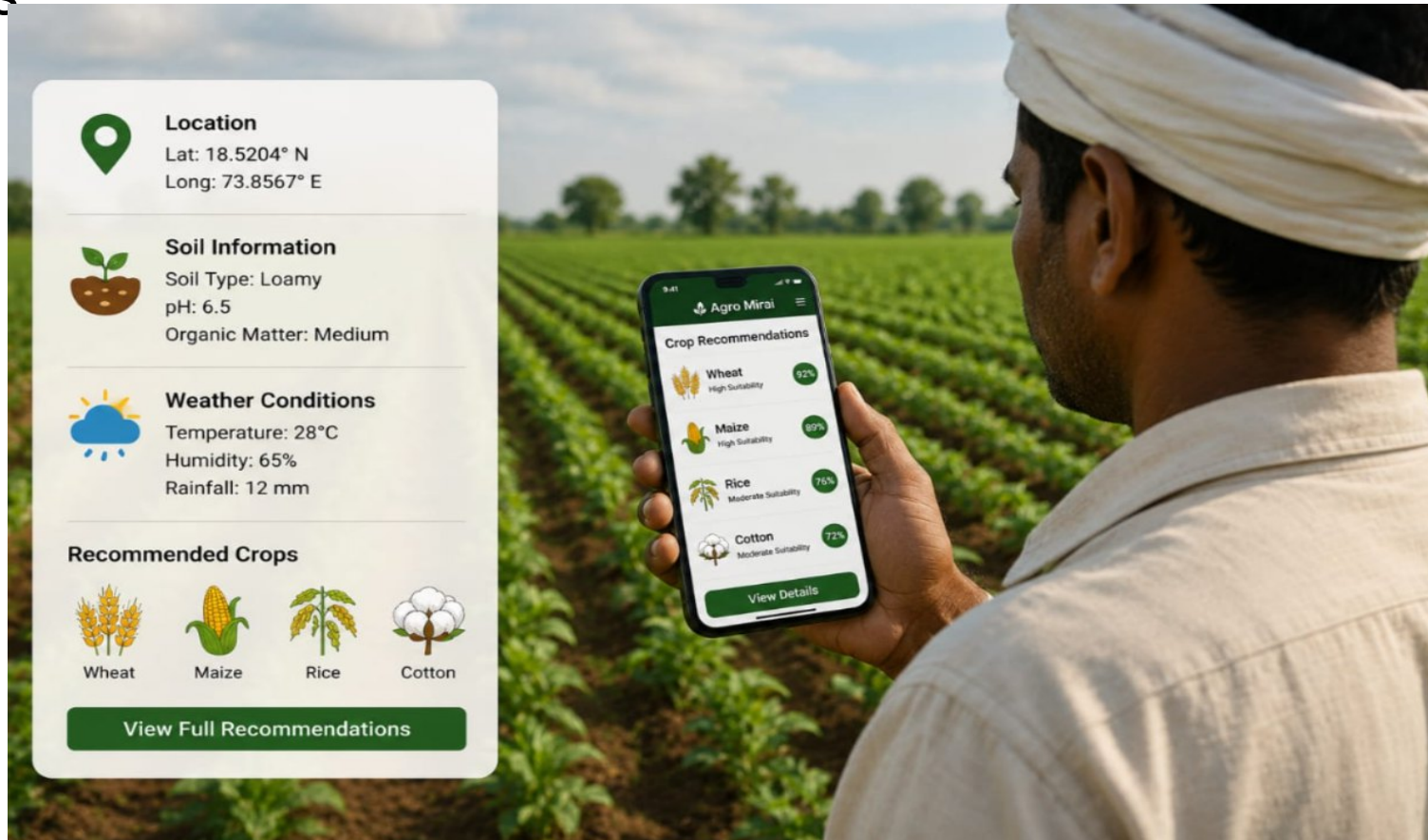


Fig 11: Expected outcome which shows crop recommendation

11. Expected Outcomes

- Improved decision-making through integration of real-time and historical data .



Fig 12: Expected outcome which shows decision making of data

11. Expected Outcomes

- Efficient irrigation planning and early crop health/disease prediction using predictive analytics .



Fig 13: Expected Outcome which shows crop recommendation

11. Expected Outcomes

- Optimized season-wise cropping patterns based on water availability and environmental factors .



Fig 14: Expected Outcome which shows season wise crop patterns

11. Expected Outcomes

- Accessible, user-friendly, multilingual, and voice-enabled system for wider farmer adoption .



Fig 15: Expected Outcome which Shows Multilingual feature

12. Applications

- **Agricultural fields** for real-time crop monitoring and smart advisory .
- **Small and large farms** for precision farming and resource optimization .
- **Rural areas** for data-driven decision support to farmers .
- **Government agricultural programs** for large-scale farm monitoring and planning .
- **Agri-tech platforms** for intelligent crop management and predictive analytics .



Fig 8: Application of AGRO MIRAI

13. Future scope

- Autonomous Farming: AI-powered robots & drones for automated monitoring.
- Edge AI: Enables full offline functionality in rural areas
- Digital Twin: Simulates farm conditions for better decision-making
- Blockchain: Ensures secure data sharing and transparent supply chains

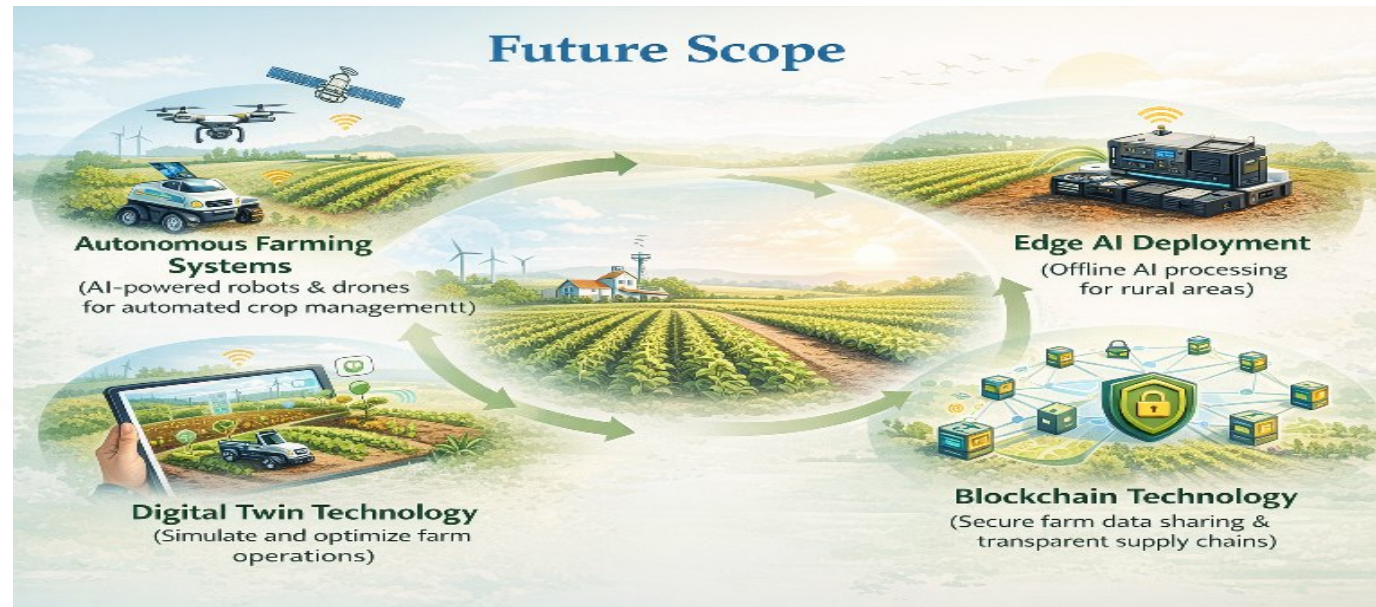


Fig 9: Future Scope of AGRO MIRAI

14. Work Plan

Phase	Description	Duration
1	Literature Survey + Problem Statement + Synopsis	6 weeks
2	Requirement Analysis + Design	5 weeks
3	Implementation (Module Development)	10 weeks
4	Integration + Testing + Deployment + Final Report	7 weeks

15. Conclusion

This project presents an AI-driven Smart Agriculture Advisory System that integrates field data, weather information, and satellite insights to support intelligent farming decisions. The system provides accurate recommendations, predictive analysis, and real-time guidance, helping farmers improve productivity and optimize resources. By enabling data-driven and proactive agriculture, the proposed solution contributes to sustainable farming and demonstrates strong potential for real-world and large-scale applications.



Fig 10: Conclusion of AGRO MIRAI

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Thank You...!!!